Judge Identity as a Confounding Factor: An Audit of LLM-as-Judge Evaluation for a Multi-Agent Discussion System

Anonymous Authors

Anonymous Submission

Multi-agent discussion systems let several role-conditioned agents deliberate on a question, and their evaluation increasingly relies on a single large language model (LLM) acting as judge: the judge reads the discussion transcript and scores quality dimensions, and the resulting scores drive claims about which system components matter. The robustness of such claims to the identity of the judge has not been examined. We audit this evaluation protocol on MADF, a decentralized roundtable framework in which each agent bids for the floor with a self-assessed confidence score. We re-implemented the framework's speaking protocol as a headless, dependency-free testbed, and scored 32 discussions (one warmup discussion excluded) with three judges drawn from two vendors: two models of the driver's vendor and one cross-vendor model. Three findings emerge. First, rank agreement between judges is low and heterogeneous (Spearman ρ from -0.10 to $+0.28$ among the three analyzed judges; $+0.47$ at most including a fourth released judge), and mean absolute differences run up to 0.87 points on a 10-point scale; in a parser-clean sensitivity analysis one cross-vendor pair reaches $\rho = +0.61$ (moderate) while level differences remain large. Second, the judges form a leniency ladder—one scores $+0.61$ above the two-judge mean and another 0.62 below it, both with bootstrap confidence intervals excluding zero, and the offsets persist on transcripts the judge did not produce—while self-preference is not detected; this refines the standard advice that using a different judge family suffices. Third, component-importance conclusions drawn from a single-judge ablation vary with the judge (Cliff's δ from -1.00 to $+0.78$ across judges and drivers), and in a same-machine replication where only the judge varies, no judge reproduces the audited evaluation's component ordering on both contrasts; no contrast survives multiplicity correction, so we report direction and effect size rather than significance. We conclude that single-judge LLM scores do not support component-importance claims about multi-agent discussion systems in their current form, and we propose a case-study-derived multi-judge reporting heuristic. All code, transcripts, judgments, a robustness re-judgment of the discussions affected by a defect we found in the audited judge prompt, and an auxiliary model review record are released with the paper.

1 Introduction

Multi-agent systems built from large language models (LLMs) are increasingly evaluated by having another LLM read the system's output and score it. In multi-

agent *discussion* systems—several role-conditioned agents deliberating on an open question—the scored artifact is the discussion transcript itself, and the scores routinely carry scientific weight: a recent open-source roundtable framework reports that its decentralized speaking mechanism outperforms single-model role-play (6.20 vs. 3.53 on a five-dimension judge scale) and ranks five system components by their measured contribution (host summary -1.20, confidence arbitration -0.92, jitter -0.80, iron rules -0.53, opening -0.40) [8]. Conclusions of this form—"this component matters more than that one"—now appear across the multi-agent literature.

All such conclusions inherit an unexamined dependency: the identity of the judge model. LLM judges are known to be imperfect measuring instruments in general [15, 10], and large-scale audits have shown that judge rankings move across benchmarks and that reliability is item-specific rather than model-level [9, 6]. What has not been measured is the consequence for the specific inferential move that multi-agent papers make: a single judge scores transcripts of system variants, and the resulting differences are read as properties of the *system*. If a different judge reverses the ordering of the variants, the component ranking is not established as a property of the system.

This paper is an audit of that inferential move, conducted on a system whose source, protocol, and original evaluation are fully available to us. We ask three research questions. **RQ1:** How consistent are different judges' quality assessments of the same multi-agent discussions? **RQ2:** Through what mechanism does judge identity act—a preference for outputs of the judge's own family, or a general leniency offset? **RQ3:** Do the component-importance conclusions of a single-judge evaluation replicate when the judge changes?

Our approach has three parts. First, we re-implemented the audited system's speaking protocol as a headless testbed with no third-party dependencies (system prompt, decision schema, confidence arbitration, jitter algorithm, and host prompts reproduced from the source; deviations listed in §3), so that every condition differs only in the factor under study. Second, we ran a 2×2 factorial experiment crossing driver model with judge model, plus a replication of the system's component ablation under multiple judges, including a same-machine replication in which only the judge varies. Third, we scored all transcripts with three judges from two vendors—two models of the driver's vendor and one cross-vendor model—so that judge identity is a controlled factor rather than an assumption.

Contributions. (i) A protocol-faithful, released testbed for multi-agent discussion evaluation (all prompts and per-round decision logs included). (ii) A quantification of cross-judge agreement in this setting: rank correlation is low and heterogeneous ($\rho \leq +0.28$ among the three analyzed judges, $+0.47$ at most including a fourth released judge; one cross-vendor pair reaches $+0.61$ in a parser-clean sensitivity analysis), and level differences reach 0.87 points on a 10-point scale. (iii) A mechanism decomposition showing that the dominant inter-judge effect is a leniency offset whose confidence interval excludes zero and that persists on transcripts the judge did not produce, while self-preference is not detected—

refining the standard mitigation advice that a different judge family suffices. (iv) A replication audit showing that a single judge's component-importance conclusions vary with the judge (Cliff's δ from -1.00 to $+0.78$ across judges and drivers), that no judge reproduces the audited evaluation's component ordering in the same-machine replication, and that no judge reproduces the audited effect magnitudes; with three-to-five repetitions per condition, no contrast survives multiplicity correction, so we report direction and effect size rather than significance.

All quantitative claims are scoped to the studied setting: one discussion framework, one ablation topic, five repetitions per ablation condition (three in the same-machine replication), and 32 discussions in total; we state these boundaries where they apply.

Scope and posture. The first author is the developer of the audited framework, and the audited evaluation is the first author's own prior work. We treat this as a case study of an evaluation protocol, not as evidence about the framework's quality: mechanism metrics (§4.5) indicate the framework operates as designed, and nothing here claims otherwise.

The rest of the paper is organized as follows. §2 positions the work against the LLM-judge reliability literature, multi-agent evaluation, and reproducibility audits. §3 describes the audited protocol, the testbed, and the experimental design. §4 reports the three findings. §5 states the minimum reporting norm we propose, and §6 concludes.

2 Related Work

Reliability of LLM judges. The LLM-as-a-judge paradigm was popularized by MT-Bench, which reported human-comparable agreement alongside systematic biases—position, verbosity, and self-enhancement [15]. Subsequent work questioned the metrics used to certify judges: inter-rater reliability between judge models ignores that judges are stochastic systems themselves [10], and exact-match agreement overstates discriminative power relative to chance-corrected measures [9]. The largest audits to date evaluate dozens of judges across hundreds of thousands of judgments, finding that judge rankings shift materially across benchmarks [9] and that reliability is item-specific rather than a model-level property [6]; agreement with human annotators has been catalogued for over fifty models [4], and self-preference bias has been quantified and mitigated in controlled designs [13]. Closest to our object of inference, JuStRank [**?**] benchmarks 48 judges as *system rankers* against a Chatbot Arena ground truth, showing that instance-level judge quality does not guarantee correct system ranking and characterizing per-judge system bias and "decisiveness". Three differences keep our measurement distinct: the ranked objects there are independent chatbot systems with different outputs, whereas ours are component variants of one system generated from *identical* transcripts; the reference is a human ground truth, whereas we have none and ask instead whether the conclusion is stable across judges; and the quantity of interest there is a system ranking, whereas ours is a component-importance

conclusion drawn from an ablation. Our setting differs in the artifact under judgment (multi-party discussion transcripts) and, critically, in the inferential use: these works ask whether judges are reliable; we ask what judge identity does to *component-importance conclusions* derived from judge scores.

Mitigations: panels and escalation. Two mitigation families are established. Panels of diverse models ("juries") aggregate judgments so that individual biases cancel [12], and escalation schemes escalate low-agreement cases to humans with statistical guarantees [5]. Both are recommendations for evaluation practice; neither has been validated for the multi-agent-discussion artifact, and neither measures what happens to published component rankings when panels replace single judges. Our audit provides exactly that measurement, and our proposed reporting norm operationalizes the escalation idea at the level of conclusions rather than items.

Evaluation of multi-agent systems. Multi-agent debate improves reasoning and factuality in task-oriented settings [2], and frameworks such as MALLM make debate components configurable with integrated evaluation pipelines [1]. Surveys of agent-as-a-judge catalog single-model, multi-agent, and agentic evaluators [14]. Closest to our object of study, *Who Speaks Next?* applies conversation-analytic norms (adjacency pairs, self-selection from internal states) to speaker control in multi-party LLM dialogue and evaluates the result with LLM judges [8]; voice-agent research learns turn-taking decisions with chunk-level ground truth and validates them with deterministic metrics plus a single auxiliary judge model [7]; and asynchronous group communication studies when an agent should speak, using human-anchored metrics [3]. None of these works varies the *judge* as an experimental factor, and none reports structural discussion metrics (speaking inequality, responsiveness) or task-outcome anchoring. Our study supplies the missing factor and the judge-free metrics.

Reproducibility audits as a genre. Audits of published results are an established publication form: reproducibility studies of evaluation pipelines [10], reporting standards for reproducible LLM evaluation, and venue-level replication programs all treat the verification of a field's methods as a first-class contribution. This paper follows that genre: the audited object is not a model's capability but an evaluation *protocol*, and the artifact under audit is the audited framework's own published-style evaluation.

3 Method

3.1 The audited system and protocol

MADF (Multi-Agent Discussion Framework) is a roundtable platform in which each participant is an LLM agent conditioned by a persona skill file. Each round proceeds as follows: every agent independently emits a decision JSON {`decision, confidence, reasoning`}; the orchestrator selects the agent with

Table 1. Positioning against prior work. "This paper" supplies the judge-identity factor, cross-vendor judging, structural (judge-free) metrics, and a same-machine replication in the multi-agent discussion setting.

Work	Setting	Judge varied	Judge-free metrics	Component audit
MT-Bench [15]	chat pref.	no	no	no
Juries [12]	generation	panel	no	no
Trust-or-escalate [5]	general	escalation	no	no
Large audits [9, 6]	benchmarks	yes	no	no
Who Speaks Next? [8]	game dialogue	no	no	no
MALLM [1]	debate	no	partial	partial
JuStRank [**?**]	chatbot systems	yes (48)	no	no
This paper	discussion	yes (3, 2 vendors)	yes	yes

the highest confidence among those choosing `speak`; if all agents wait, one is chosen at random (forced speech). Confidence values pass through a hash-derived jitter that preserves the tenths digit and replaces the hundredths digit with a value derived from the agent's run identifier, preventing the model's preference for round numbers. A host agent opens and closes the discussion. We audited commit `2f1baa`. The bidding-for-the-floor idea descends from importance-driven turn-bidding in spoken dialogue systems [11]; our audit concerns the evaluation protocol, not the mechanism's novelty.

3.2 Testbed: protocol-faithful reimplementation

The reference implementation depends on an agent-graph framework, a database, and a cache. To make every condition differ only in the factor under study, we re-implemented the protocol as a headless testbed in pure Python (standard library only). The system prompt (environment description, persona framing, three iron rules, thinking and speaking instructions, information-boundary rules), the decision schema, the jitter algorithm, the arbitration rule, and the host prompts are reproduced from the source. Documented deviations, applied identically to all conditions and therefore invisible to comparisons: the agent-graph is replaced by direct system-prompt calls (the reference implementation already bypasses the graph for decisions); streaming is replaced by single calls; persistence is omitted; persona skills are authored by us because the reference implementation's skill volumes are not in the repository; discussion duration is 120 s with a floor of three rounds. Fidelity was checked line-by-line against the source; the full correspondence table is in the released repository. We therefore describe the testbed as *protocol-faithful* rather than verbatim: the orchestration logic is reproduced exactly, while the surrounding stack and the persona files are substitutions, and all conclusions are scoped to the authored skills.

3.3 Reasoning-model adaptation

The protocol was designed for non-reasoning models. Running it unmodified on a reasoning model (our primary driver) fails *silently*: chain-of-thought consumes

the token budget so that the decision JSON never appears, and the parser degrades the decision to `wait` (all agents appear silent; every round becomes forced speech); speech calls emit meta-narration instead of character speech; and judge calls emit no parseable score, defaulting to 5.0. We fixed this with per-call token budgets and reasoning-effort control, and parse decisions from both output channels. The failure is not fully eliminated, only bounded: across the 480 judged dimension calls, 28–30 per judge still required regex fallback, and 22–26 per judge still defaulted to 5.0 (Appendix B). We report this because any replication of this protocol family on reasoning models must handle it, and because silent degradation is itself a hazard for the evaluation protocols under audit.

3.4 Experimental design

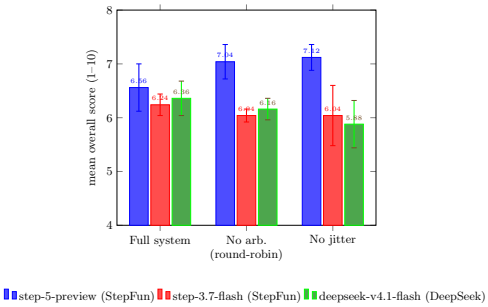
Judges. Three judges score every transcript: two models of the driver's vendor (a flagship reasoning model and a smaller non-reasoning model) and one cross-vendor model (DeepSeek `v4.1-flash`). All judges receive the audited system's own judge prompts verbatim (five dimensions, 1–10, anchored rubrics, pre-computed speaking-count statistics); only the judge model varies. Temperature 0.1. A fourth judge (a third flash-class model of the driver's vendor) was also run and is released with the artifact; it is reported only as a robustness line, because its parse-failure rate is roughly three times the others' (Appendix B).

Discussions. We analyze 32 discussions; one warmup discussion present in the released transcript file is excluded from all analyses. (i) *Factorial*: two drivers × two topics × two repetitions (8 discussions, full system) for the judge-identity analysis. (ii) *Ablation replication*: the audited system's three core conditions—full system, no arbitration (round-robin), no jitter—each five repetitions (15 discussions) under the primary driver. (iii) *Same-machine replication*: the same three conditions, three repetitions each (9 discussions), under the secondary driver, so that three judge identities—the driver's own model, a same-vendor sibling, and a cross-vendor model—can be compared on identical transcripts. Personas are fixed across conditions (three public figures with disjoint expertise); topics are fixed across conditions.

A defect in the audited judge prompt. The audited judge's pre-computed speaking-count statistic counts only agents that appear in the transcript, so an agent that never speaks is omitted and the reported maximum gap is underestimated: for counts $(2, 1, 0)$ it reports gap 1 (applying the "mechanical rotation" cap) instead of the true gap 2. This affects 7 of the 32 discussions. We preserve the audited behavior in the main analysis—the prompt is part of the protocol under audit, and the defect applies identically to every judge—and re-judge the affected discussions with corrected statistics as a robustness check (Appendix C).

Mechanism metrics. Judge-free structural metrics computed deterministically from transcripts: speaking-count Gini coefficient (over all three agents, zero-speech agents included) and forced-speech rate. These describe discussion dynamics without invoking any judge.

Fig. 1. Ablation under three judges (primary driver, five repetitions; error bars: bootstrap 95% CI). Condition ordering varies across judges: the same-model judge ranks the full system below both ablations, the cross-vendor judge above both.



3.5 Statistics

Inter-judge agreement uses Spearman rank correlation (overall and per dimension) and mean absolute difference. Condition comparisons report the mean difference, Cliff's δ (non-parametric effect size), bootstrap 95% confidence intervals for the mean difference and for δ (10,000 resamples, groups resampled independently), and an exact two-sided permutation test on the difference of means (all $\binom{n_1+n_2}{n_1}$ labelings; exact in the presence of ties). The six ablation contrasts per phase (three judges $\times$ two ablations) form the family for Holm correction; the correction was applied after the audit revisions reported in this paper, so it is stated as post hoc family-wise control rather than preregistered analysis. With three-to-five repetitions per condition the exact test has limited resolution—with 3×3 samples its minimum attainable two-sided p is 0.1—and after Holm correction no ablation contrast is significant; we therefore treat point estimates, effect sizes, and intervals as the primary evidence and p-values as auxiliary, and we make no significance claims for any single contrast. Self-preference is estimated as the difference between same-driver and other-driver scores under a fixed judge, with a stratified bootstrap interval (groups resampled independently, sizes preserved) and an exact permutation p-value over all $\binom{8}{4}$ labelings; this is an exploratory, unblocked analysis on four versus four discussions and is underpowered. Leniency offsets are differences from the mean of the other judges on the same discussion, bootstrapped over discussions. An auxiliary model (different family from the authors' toolchain; not a blind review) reviewed the evidence-to-claim mapping; its verdict and our responses are in Appendix A.

4 Results

4.1 Inter-judge agreement is low and heterogeneous (RQ1)

Table 2 reports pairwise agreement between the three judges on the 32 experimental discussions (overall score). No judge pair reaches strong rank agreement:

Fig. 2. Per-discussion overall scores for all three judge pairs (32 discussions). Dashed line: perfect agreement. No pair shows tight agreement.

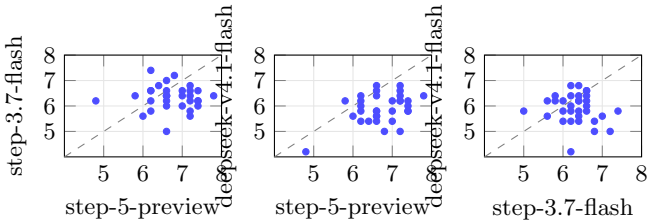
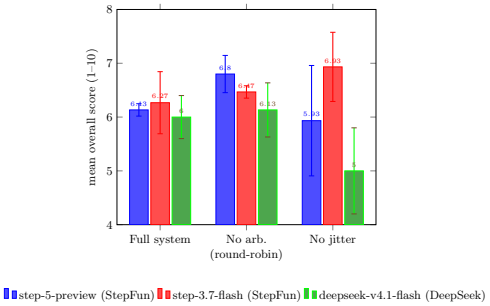


Fig. 3. Same-machine replication (secondary driver, three repetitions; error bars: SD). Identical transcripts; only the judge differs. The driver's own model, a same-vendor sibling, and a cross-vendor judge produce three different condition orderings.



among the three analyzed judges ρ spans -0.10 to $+0.28$ and mean absolute differences run from 0.66 to 0.87 points on a 10-point scale, so two judges looking at the same transcript typically differ by most of a scale point, and their rankings can be uncorrelated or slightly negatively correlated. The cross-vendor pairs behave no better than the same-vendor pair, so the disagreement is not a within-vendor artifact. A fourth released judge (a flash-class model of the driver's vendor, not analyzed in the main text) reaches $\rho = +0.47$ with one of the three—still far from strong agreement—and $+0.21/+0.26$ with the other two. Per dimension, agreement on insight depth—the most subjective axis—is negative for the two pairs involving the flagship judge (-0.23, -0.10) and only $+0.33$ for the remaining pair: the judges systematically disagree about which discussions are deep. A parser-clean sensitivity analysis (Appendix B) leaves one pair at $\rho = +0.61$ while the other two stay near zero, so we report agreement as heterogeneous rather than uniformly low. Because all judges are LLMs from two vendors, these figures describe disagreement across model families within and between two vendors; we do not claim they bound disagreement with human judges.

Fig. 4. Speaking-count inequality (Gini over all three agents, zero-speech agents included) per ablation condition, both drivers.

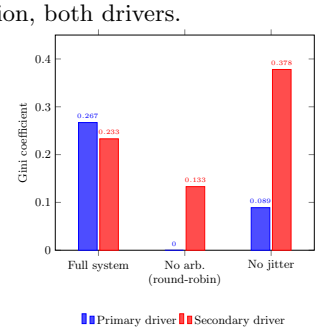


Table 2. Pairwise inter-judge agreement (overall score, 32 discussions). "Cross" marks pairs from different vendors. The released fourth judge is shown for completeness only.

Judge pair	n	Spearman ρ	Mean $\lvert\Delta\rvert$	Depth ρ
step-5-preview vs. step-3.7-flash	32	-0.05	0.72	-0.23
step-5-preview vs. deepseek-v4.1-flash (cross)	32	$+0.28$	0.87	-0.10
step-3.7-flash vs. deepseek-v4.1-flash (cross)	32	-0.10	0.66	$+0.33$
step-3.5-flash (released) vs. step-5-preview	32	$+0.21$	0.90	—
step-3.5-flash (released) vs. step-3.7-flash	32	$+0.47$	0.54	—
step-3.5-flash (released) vs. deepseek-v4.1-flash	32	$+0.26$	0.57	—

4.2 Leniency offset, not self-preference (RQ2)

The three judges form a leniency ladder (Table 3): step-5-preview scores $+0.61$ above the mean of the other two judges on the same discussion (bootstrap 95% CI $[+0.40, +0.80]$, excluding zero), deepseek-v4.1-flash scores -0.62 below it (CI $[-0.83, -0.42]$, excluding zero), and step-3.7-flash sits at the midpoint ($+0.02$, CI $[-0.23, +0.26]$). The two extreme judges thus differ by about 1.23 points on the 10-point scale for identical transcripts. Crucially, the offsets persist in cells where the judge did *not* produce the discussion: restricted to transcripts driven by another model, the offsets are $+0.38$ (CI $[+0.08, +0.68]$, $n = 13$), -0.30 (CI $[-0.55, -0.03]$, $n = 19$), and -0.62 (CI $[-0.83, -0.42]$, $n = 32$). The offsets are therefore properties of the judges, not preferences for their own family. Direct self-preference estimates are small and statistically indistinguishable from zero: $+0.05$ (stratified bootstrap CI $[-0.60, +0.75]$; exact permutation $p = 1.0$) and -0.10 (CI $[-0.50, +0.30]$; $p = 0.83$) for the two judges that also served as drivers, on four same-driver and four other-driver discussions each (an exploratory, unblocked comparison). This decomposes the two mechanisms that the standard mitigation ("use a different family") conflates: changing family removes self-preference, which was not the operative effect here, while leaving the leniency offset untouched. The non-detection of self-preference is a power statement (four vs. four discussions), not evidence of absence.

Table 3. Leniency offset per judge (score minus the mean of the other judges on the same discussion; bootstrap 95% interval, 10,000 resamples over discussions) and self-preference delta (same-driver minus other-driver discussions under that judge; stratified bootstrap interval and exact permutation p over all 70 labelings).

Judge	Leniency offset	95% CI	Self-pref. Δ (CI; p)
step-5-preview	+0.61	[+0.40, +0.80]	+0.05 ([−0.60, +0.75]; $p = 1.0$)
step-3.7-flash	+0.02	[−0.23, +0.26]	−0.10 ([−0.50, +0.30]; $p = 0.83$)
deepseek-v4.1-flash	−0.62	[−0.83, −0.42]	—

Table 4. Ablation replication, five repetitions per condition, primary driver (mean overall ± SD). δ is Cliff's delta for full vs. ablated condition under the stated judge, with exact two-sided permutation p and Holm-adjusted p^* across the six contrasts (3 judges × 2 ablations; post hoc family-wise control).

	Same-model judge		Same-vendor judge		Cross-vendor judge	
Condition	Score	δ (p)	Score	δ (p)	Score	δ (p)
Full system	6.56 ± .57	—	6.24 ± .26	—	6.36 ± .43	—
No arb. (round-robin)	7.04 ± .41	−0.48 (0.20)	6.04 ± .17	+0.56 (0.29)	6.16 ± .26	+0.32 (0.48)
No jitter	7.12 ± .30	−0.64 (0.11)	6.04 ± .74	+0.08 (0.68)	5.88 ± .59	+0.60 (0.21)

Judge identities: same-model = step-5-preview (also the driver), same-vendor = step-3.7-flash, cross-vendor = deepseek-v4.1-flash. Holm-adjusted $p^* \geq 0.67$ for all six contrasts.

4.3 Single-judge component conclusions do not replicate (RQ3)

Table 4 reports the ablation replication at five repetitions per condition under the primary driver. The audited evaluation claims that arbitration and jitter contribute −0.92 and −0.80 (removing either component *lowers* the score), i.e. the full system should outscore both ablations. The same-model judge—the judge is the driver, the single-judge protocol structure—ranks the full system *below* both ablations (Cliff's $\delta = -0.48$ and -0.64): removing components *raises* the score, the opposite of the audited direction. The cross-vendor judge runs in the audited direction on both contrasts ($\delta = +0.32$ and $+0.60$) and the same-vendor sibling on one ($+0.56$; $+0.08$ on the other). The audited magnitudes are reproduced by *no* judge at anything like that scale: the largest effect any judge reports is $|\delta| = 0.64$, and it points the other way. With five repetitions per condition no contrast is significant after Holm correction across the six contrasts (three judges × two ablations; correction applied post hoc; raw exact p from 0.11 to 0.68, adjusted $p \geq 0.67$), so we read this table as *direction and effect size*, not as a significance test: the finding is that the direction of the component effect is a property of the judge, not that any single contrast is established.

Table 5. Same-machine replication (secondary driver, three repetitions per condition, mean overall ± SD). Identical transcripts; only the judge differs. δ is Cliff's delta for full vs. ablated condition.

	Driver's own model		Same-vendor judge		Cross-vendor judge	
Condition	Score	δ	Score	δ	Score	δ
Full system	$6.27 \pm .58$	—	$6.13 \pm .12$	—	$6.00 \pm .40$	—
No arb. (round-robin)	$6.47 \pm .12$	$+0.11$	$6.80 \pm .35$	-1.00	$6.13 \pm .50$	-0.22
No jitter	$6.93 \pm .64$	-0.56	5.93 ± 1.03	-0.11	$5.00 \pm .80$	$+0.78$

Judge identities: driver's own model = step-3.7-flash (also the driver), same-vendor = step-5-preview, cross-vendor = deepseek-v4.1-flash. Exact permutation p-values: 1.0, 0.1, 1.0 (own model); 0.1, 1.0, 1.0 (same-vendor); 0.8, 0.2, 1.0 (cross-vendor), for round-robin/no-jitter contrasts as listed.

4.4 Same-machine replication: only the judge varies

Table 5 holds transcripts fixed and varies only the judge, on the secondary driver (three repetitions per condition). The three judge identities produce three different condition orderings on *identical* transcripts: the driver's own model ranks the full system below the no-jitter ablation ($\delta = -0.56$); the same-vendor sibling ranks it below the round-robin ablation at the maximum effect size ($\delta = -1.00$: every round-robin discussion outscores every full-system discussion, exact $p = 0.1$, the resolution floor at 3×3); and the cross-vendor judge ranks the full system above the no-jitter ablation ($\delta = +0.78$; interval for the mean difference $[+0.20, +1.80]$ excludes zero, but exact $p = 0.2$ and Holm-adjusted $p = 1.0$, so we treat it as suggestive only). Across both drivers and all three judges, no judge–driver combination reproduces the audited evaluation's component ordering on both contrasts. This is the strongest form of the evidence available at this sample size: the transcripts are held fixed, so the only changed factor is the judge, and the ranking moves. We do not claim a driver-independent causal effect beyond the tested transcripts; the factorial's driver and judge factors are partially confounded by design (§5).

4.5 Mechanism metrics: the framework operates as designed

Judge-free metrics confirm the ablations manipulate what they claim (Table 6). Under the primary driver, the full system produces unequal speaking counts (Gini = 0.27), round-robin produces a fixed rotation (Gini = 0.00), and removing jitter collapses most of the asymmetry (Gini = 0.09), consistent with jitter as the operative source of speaking inequality under this driver. Under the secondary driver the pattern does not hold (no-jitter Gini = 0.38 exceeds the full system's 0.23), so we do not generalize the jitter claim beyond the primary driver. Forced speech never occurs under round-robin (by construction) and occurs at similar rates under the other two conditions. Note that these Gini values count all three agents, with zero-speech agents included; the audited pipeline's own turn-count

Table 6. Mechanism metrics per ablation condition (deterministic, no judge involved). Gini: speaking-count inequality over all three agents; forced: rounds in which all agents waited and a speaker was forced, over the actual number of rounds.

	Primary driver		Secondary driver	
Condition	Gini	Forced	Gini	Forced
Full system	0.27	5/15	0.23	1/14
No arbitration	0.00	0/15	0.13	0/15
No jitter	0.09	5/15	0.38	1/14

statistic omits zero-speech agents (§3), which is the defect whose robustness check is reported in Appendix C. The framework's speaking mechanism behaves as documented under the primary driver; the instability documented above resides in the evaluation layer, not the framework.

5 Discussion

A case-study-derived reporting heuristic. Our results support a concrete minimum for papers that derive component-importance claims from LLM-judged multi-agent artifacts: (1) score with at least two judges from different model families, and preferably different vendors; (2) report pairwise judge agreement and per-judge offsets alongside the scores; (3) when judges disagree on the ordering of the variants, downgrade the component claim to "undetermined" or escalate to human judgment, rather than reporting the single judge's p-value. Condition (3) is the escalation idea of [5] applied at the level of conclusions, and (1)–(2) are the panel idea of [12] reduced to its measurement core. We offer this as a heuristic derived from one case study: what the study establishes is the need to report judge sensitivity, not that this particular panel size is sufficient.

What the leniency result changes. The community's default mitigation for judge bias is family substitution. Our decomposition shows why that is insufficient here: the operative effect is a between-judge level shift (offset interval excluding zero, persisting on transcripts the judge did not produce) that survives family substitution, while self-preference—the effect family substitution targets—is not detected at the available power. A judge panel with a level-calibration step (e.g., anchoring all judges to a common reference set) is the natural repair; we leave its validation to future work.

Statistical honesty at this sample size. With five repetitions per ablation condition (three in the same-machine replication), no contrast survives multiplicity correction, and the 3×3 exact test cannot return $p < 0.1$ at all. We therefore make no significance claim for any single contrast. The evidence for RQ3 is structural: on identical transcripts, changing only the judge changes the direction of the component effect, with effect sizes up to $|\delta| = 1.00$; and the audited evaluation's

reported magnitudes (-0.92, -0.80) are not approached by any judge we tested. Direction reversal under a controlled factor is a weaker but more robust form of evidence than a p-value at this n, and it is sufficient to break the specific inferential move under audit: reading a single judge's component differences as properties of the system.

Relationship to general judge-reliability findings. That LLM judges are imperfect is established [9, 6]. The contribution here is downstream: measuring what that imperfection does to a specific scientific inference (component ranking), in a setting where no ground truth anchors the scores, and separating two mechanisms that prior work treats jointly. The same-machine replication holds transcripts fixed so that judge identity is the only changed factor within the tested set; because the factorial's driver and judge factors are partially confounded by design, we do not claim a driver-independent causal effect beyond the tested transcripts.

A defect found in the audited judge. The audit surfaced a concrete defect in the audited evaluation's judge prompt: the pre-computed speaking-count statistic omits agents that never speak, understating the maximum gap and misapplying the prompt's own hard caps in 7 of 32 discussions. We preserved the audited behavior in the main analysis (the prompt is part of the protocol under audit, and the defect applies identically to every judge, so it cannot manufacture inter-judge disagreement) and verified by re-judgment that the affected discussions do not carry the qualitative conclusions (Appendix C). Reporting the defect is part of the audit's value: it is exactly the kind of silent evaluation bug that single-judge protocols do not surface.

6 Conclusion

We audited the evaluation protocol by which a multi-agent discussion system's component importance is established. Re-implementing the speaking protocol as a headless testbed and scoring 32 discussions with three judges from two vendors, we found that no judge pair reaches strong rank agreement on discussion quality ($\rho \leq +0.28$ among the analyzed judges); that the judges form a leniency ladder whose extremes differ by about 1.23 points on a 10-point scale, persisting on transcripts the judges did not produce, while self-preference is not detected—refining the standard family-substitution mitigation; and that a single judge's component-importance conclusions vary with the judge (Cliff's δ from -1.00 to $+0.78$), with no judge reproducing the audited evaluation's component ordering in the same-machine replication and no judge reproducing its effect magnitudes. With the available repetitions no contrast is statistically significant after multiplicity correction; the finding is the instability of the direction, not a significant difference.

The practical upshot is a reporting norm rather than a prohibition: multi-judge scoring with agreement reporting, and conclusion-level escalation when judges disagree.

Limitations and next steps. Four limitations bound these claims, each with the experiment that would relax it. (i) *Scope*: one framework and, for the ablation, one topic; a multi-topic, multi-framework replication is the direct next step. (ii) *Statistical power*: five repetitions per ablation condition (three in the same-machine replication) limit resolution, and LLM sampling adds run-to-run variance; larger repetition counts and repeated runs at fixed seeds would tighten every interval. (iii) *No human calibration*: we show that judges disagree with each other, not which judge (if any) is right; human annotation of a transcript subset is the calibration step and is the backbone of our companion study on speaking-allocation policies. (iv) *Judge provenance*: the judges span two vendors but include only one cross-vendor model, and the audited framework's own evaluation used a third vendor's judge, so our figures describe disagreement across a two-vendor slice of model space; a wider judge panel is a stated extension of the released pipeline. The judge-free mechanism metrics and the released artifacts (code, transcripts, decision logs, judgments, robustness re-judgment, jury record) make every number in this paper checkable. Immediate next steps are a wider judge panel at scale, human-annotation calibration of the judge-free metrics, and a multi-topic expansion—the last two are the substance of our companion study on speaking-allocation policies, where these metrics serve as the evaluation backbone.

Ethics and reproducibility. The first author developed the audited framework; the audit targets the evaluation protocol, and the repository contains the protocol-fidelity mapping, all prompts, all transcripts with per-round decision logs, all raw judgments, and the analysis scripts. No human subjects were involved; all agents are LLM-driven with public-persona skill files.

A Auxiliary model review record

Because the authors audited their own system's evaluation, an independent model (different family from the authors' toolchain) reviewed the evidence-to-claim mapping before this manuscript was finalized; this is an auxiliary review by a model outside the authors' toolchain, not a blind review and not an independent replication. The jury received the full results, the candidate claims, the known caveats, and instructions to avoid inflating claims beyond the data. Its verdict was *partial* support with *medium* confidence. Points the jury raised and how this manuscript responds:

- *"Non-significance under small n is a design property, not a finding."* Accepted: the manuscript's primary evidence for RQ3 is the direction reversal with maximum effect size and intervals for the differences; exact p-values are reported as auxiliary and no significance claim is made for any single contrast.
- *"Self-preference non-detection is a power statement, not proof of absence."* Accepted: claims are stated as "not detected" with intervals and an exact permutation p-value, never as absence.

– *"Driver capability is partly confounded with judge identity in the factorial."* Accepted: the same-machine replication (§4) holds transcripts fixed and varies only the judge, the confounding is stated in §5, and no driver-independent causal claim is made beyond the tested transcripts.
– *"The setting is narrow."* Accepted: the posture is a protocol audit case study; generality is claimed for the inferential move (single-judge component ranking), not for the framework.

The full jury transcript, the deterministic evidence pre-check, and the claim-by-claim responses are released with the repository.

B Judge data quality

Table 7 quantifies the reasoning-model degradation described in §3 per judge. "Parse-fallback" counts dimension calls whose output required regex extraction instead of direct JSON parsing; "defaulted" counts dimension calls that ended at the 5.0 fallback (a subset of parse-fallback: no recoverable score); "direct" counts calls parsed directly from JSON. The released fourth judge's degradation rate is roughly three times the others', which is why it is excluded from the main analysis. No judgment is excluded from any analysis on these grounds; the counts are reported so readers can weigh the evidence.

Table 7. Per-judge data quality over the 32 analyzed discussions (160 dimension calls each).

Judge	Mean overall	Direct JSON	Parse-fallback	Defaulted to 5.0
step-5-preview	6.73	132/160	28/160	24/160
step-3.7-flash	6.33	133/160	27/160	22/160
deepseek-v4.1-flash	5.91	130/160	30/160	26/160
step-3.5-flash (released only)	5.93	82/160	78/160	59/160

Parser-robustness sensitivity. Because the 5.0 default enters the reported overall scores, we re-ran the headline analyses on a parser-clean subset: parse-fallback dimensions treated as missing, a discussion–judge cell kept when at least three of five dimensions were directly parsed (30, 29, and 30 of 32 cells retained per judge; script and output released). The clean overall is a complete-case estimand (the mean of the directly parsed dimensions, at least three of five), not a corrected estimate of the five-dimension score; the comparison quantifies sensitivity, not validation. The leniency ladder is preserved, with intervals still excluding zero for the two extremes ($+0.63$ $[+0.46, +0.79]$, $+0.03$ $[-0.27, +0.34]$, -0.66 $[-0.86, -0.47]$). The same-model judge's direction reversal (Table 4) is preserved on both contrasts. One of the six p2 direction labels moves—the cross-vendor judge's no-jitter cell flips to full-below with only three of five discussions

retained—so direction claims for that judge are treated as suggestive. Rank agreement remains low for two pairs ($\rho = +0.03$, -0.09) but one cross-vendor pair rises from $+0.28$ to $+0.61$ (moderate) on the clean subset, while mean absolute differences stay large (0.73–0.84). We therefore state the agreement finding as heterogeneous and modest rather than uniformly near-zero, and the sensitivity numbers above show how much the parser fallback moves each claim.

C Robustness re-judgment of the turn-count defect

The audited judge prompt's pre-computed speaking-count statistic omits agents that never speak (§3), affecting 7 of the 32 discussions: `p1-step-5-preview-t2-r1`, `p2-full-r1/r2/r3`, `p2-no_jitter-r3`, `p3-no_jitter-r1/r3`. We re-judged exactly these discussions with corrected statistics (all three agents counted, true maximum gap), under the same three judges, prompts, and temperature. Table 8 reports the effect. The corrected re-judgment did not overturn any qualitative direction: under corrected statistics the same-model judge still ranks the full system below both ablations (cell means 6.96 vs. 7.04 and 7.40), the cross-vendor judge still ranks it above both (6.68 vs. 6.16 and 6.04), and the same-machine orderings are unchanged. Because the re-judgment is a fresh sample at temperature 0.1, the observed score shifts—including the +1.00 average rise on the flagship judge's previously capped dimensions—cannot be attributed solely to the prompt correction; they bound the defect's plausible influence rather than identifying its causal effect. The main analysis therefore keeps the audited prompts, and this appendix establishes that the reported directions survive the correction. (A separate sensitivity analysis on the parser-clean subset, §B, moves one cross-vendor direction label; both checks are reported so neither is read as stronger than it is.)

Table 8. Robustness of the 7 affected discussions to the corrected turn-count statistic. "Mean shift": corrected minus audited overall, averaged over the 7 discussions. "Capped dims": shift on coherence and engagement (14 dimension calls per judge). Cell means: p2 full-system and no-jitter cells with the affected repetitions re-judged.

Judge	Mean shift	Capped dims	p2 full	p2 no-jitter	Direction
step-5-preview	+0.51	+1.00	6.56 → 6.96	7.12 → 7.40	kept
step-3.7-flash	−0.20	+0.07	6.24 → 6.04	6.04 → 5.92	kept
deepseek-v4.1-flash	+0.26	+0.14	6.36 → 6.68	5.88 → 6.04	kept

D Artifact contents

All paths below are relative to `exam2/` in the released repository.

– `madf_harness.py` — headless testbed (protocol reimplementation, prompts and algorithms reproduced from the audited source; fidelity mapping included).

- `run_experiment.py`, `score_third_judge.py`, `rejudge_robustness.py` — experiment, multi-judge scoring, and robustness re-judgment runners.
- `final_analysis_v2.py`, `make_tikz_figures_v2.py`, `parser_sensitivity.py` — analysis, figure generation, and the parser-robustness sensitivity check (every number and figure in the paper is reproducible from these scripts plus the raw files below).
- `results/raw/discussions.jsonl` — every discussion with full transcript and per-round decision logs (decision, confidence, reasoning, jittered confidence, speaker, forced flag); includes the excluded warmup discussion, labeled by job id.
- `results/raw/judgments.jsonl` — every raw judge output per dimension, for the three analyzed judges and the released fourth judge.
- `results/raw/judgments_robustness.jsonl` — the corrected-statistics re-judgment of the 7 affected discussions.
- `results/analysis/final_stats_v2.json` — all reported statistics with intervals and exact p-values.
- `results/analysis/parser_sensitivity.json` — the parser-clean sensitivity analysis.
- `CLAIMS_FROM_RESULTS.md` — jury verdict and responses; `.aris/traces/` — jury transcripts.

Bibliography

[1] Becker, J., Kaesberg, L.B., Bauer, N., Wahle, J.P., Ruas, T., Gipp, B.: MALLM: Multi-agent large language models framework. In: Habernal, I., Schulam, P., Tiedemann, J. (eds.) Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. pp. 418–439. Association for Computational Linguistics, Suzhou, China (Nov 2025). https://doi.org/10.18653/v1/2025.emnlp-demos.29, https://aclanthology.org/2025.emnlp-demos.29/

[2] Du, Y., Li, S., Torralba, A., Tenenbaum, J.B., Mordatch, I.: Improving factuality and reasoning in language models through multiagent debate (2023). https://doi.org/10.48550/ARXIV.2305.14325, https://arxiv.org/abs/2305.14325

[3] Eckhaus, N., Berger, U., Stanovsky, G.: Time to talk: LLM agents for asynchronous group communication in mafia games. In: Christodoulopoulos, C., Chakraborty, T., Rose, C., Peng, V. (eds.) Findings of the Association for Computational Linguistics: EMNLP 2025. pp. 11356–11368. Association for Computational Linguistics, Suzhou, China (Nov 2025). https://doi.org/10.18653/v1/2025.findings-emnlp.608, https://aclanthology.org/2025.findings-emnlp.608/

[4] Han, S., Junior, G.T., Balough, T., Zhou, W.: Judge's verdict: A comprehensive analysis of llm judge capability through human agreement (2025). https://doi.org/10.48550/ARXIV.2510.09738, https://arxiv.org/abs/2510.09738

[5] Jung, J., Brahman, F., Choi, Y.: Trust or escalate: Llm judges with provable guarantees for human agreement (2024). https://doi.org/10.48550/ARXIV.2407.18370, https://arxiv.org/abs/2407.18370

[6] Kumar, V., Rathore, D., Moitra, A.: All verdicts are not equal: Rethinking llm judge reliability (2026). https://doi.org/10.48550/ARXIV.2610.12083, https://arxiv.org/abs/2610.12083

[7] Mitra, S., Pandey, P., Jain, A., Sahith, S., Girish, K.V.V.: Adaptive turn-taking for real-time multi-party voice agents (2026). https://doi.org/10.48550/ARXIV.2606.13544, https://arxiv.org/abs/2606.13544

[8] Nonomura, R., Mori, H.: Who speaks next? multi-party ai discussion leveraging the systematics of turn-taking in murder mystery games (2024). https://doi.org/10.48550/ARXIV.2412.04937, https://arxiv.org/abs/2412.04937

[9] Norman, J.D., Rivera, M.U., Hughes, D.A.: Reliability without validity: A systematic, large-scale evaluation of llm-as-a-judge models across agreement, consistency, and bias (2026). https://doi.org/10.48550/ARXIV.2606.19544, https://arxiv.org/abs/2606.19544

[10] Schroeder, K., Wood-Doughty, Z.: Can you trust llm judgments? reliability of llm-as-a-judge (2024). https://doi.org/10.48550/ARXIV.2412.12509, https://arxiv.org/abs/2412.12509

[11] Selfridge, E., Heeman, P.: Importance-driven turn-bidding for spoken dialogue systems. In: Hajič, J., Carberry, S., Clark, S., Nivre, J. (eds.) Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics. pp. 177–185. Association for Computational Linguistics, Uppsala, Sweden (Jul 2010), https://aclanthology.org/P10-1019/

[12] Verga, P., Hofstatter, S., Althammer, S., Su, Y., Piktus, A., Arkhangorodsky, A., Xu, M., White, N., Lewis, P.: Replacing judges with juries: Evaluating llm generations with a panel of diverse models (2024). https://doi.org/10.48550/ARXIV.2404.18796, https://arxiv.org/abs/2404.18796

[13] Yang, J., Hu, Z., Qiu, C., Deng, Z., Jiao, X., Zhou, T.: Quantifying and mitigating self-preference bias of llm judges (2026). https://doi.org/10.48550/ARXIV.2604.22891, https://arxiv.org/abs/2604.22891

[14] Yu, F.: When ais judge ais: The rise of agent-as-a-judge evaluation for llms (2025). https://doi.org/10.48550/ARXIV.2508.02994, https://arxiv.org/abs/2508.02994

[15] Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging llm-as-a-judge with mt-bench and chatbot arena (2023). https://doi.org/10.48550/ARXIV.2306.05685, https://arxiv.org/abs/2306.05685